Excel Automation & Data Processing Services

I build tools that replace manual spreadsheet work with reliable, repeatable processes

Darek Krawczynski | ytcodes.work@gmail.com | Salt Lake City, UT

What Manual Spreadsheet Work Actually Costs

Most teams know their processes could be faster. The real cost is not just the hours. It is the inconsistency between runs, the errors that only surface downstream, and the institutional knowledge that lives in one person's head instead of in a tool anyone can run.

3-5 hrs/wk

spent on tasks that repeat the same steps every time

15-30%

of manual entries contain formatting or data errors

1 person

often holds the entire process in their head

The four capabilities on the following slides are the building blocks I use to solve these problems. Each one is drawn from tools I have built and deployed in real business environments.

Data Extraction & Validation

Pull clean, validated data from any source file into a structured format

What this means for your business

Your team receives files from vendors, banks, ERP exports, or internal departments in different formats every cycle. Columns shift, headers change, date formats vary, prices arrive in European or US notation. Instead of manually fixing each file, a tool reads the source, detects the layout, validates every row against your business rules, and writes a clean output ready for the next step.

Example: I built a tool that reads a yearly RFQ spreadsheet, validates BPA numbers, resolves buyer codes, normalises prices from multiple currency formats, and produces a validated dataset in under 30 seconds. The manual version took 45-60 minutes per batch.

If you implement this

- A weekly or monthly data preparation task that takes 30-60 minutes drops to under a minute, with every row validated before it leaves the tool.
- Validation catches problems at the source. Missing fields, invalid codes, and format errors are flagged in a log sheet before they reach your downstream system.
- The tool adapts when your source file layout changes. Column mappings are configurable, so a new vendor format or a revised ERP export does not require a rebuild.

Record Matching & Deduplication

Find matching records across systems even when names, codes, and formats differ

What this means for your business

When your data comes from two or more systems, exact lookups fail. One system stores 'ACME CORP', the other stores 'Acme Corporation Pty Ltd'. Bank account numbers have different formatting. Vendor IDs have leading zeros in one system and not the other. A matching engine normalises both sides, scores similarity, and resolves the best match automatically.

Example: I built an engine that matches vendor names from bank payment files against an Oracle registry of 50,000+ accounts using Levenshtein similarity scoring with configurable thresholds. It resolves matches that exact lookups miss, including abbreviations, punctuation differences, and word-order variations.

If you implement this

- Cross-system reconciliation that currently requires manual row-by-row comparison becomes an automated pass. A batch of 200-500 records resolves in 2-3 minutes.
- Near-duplicate detection surfaces records that look different but refer to the same entity. This directly reduces duplicate payments, duplicate customer records, and duplicate vendor setups.
- The matching threshold is configurable. You control the trade-off between catching more matches and avoiding false positives, tuned to your data.

Automated File & Report Generation

Turn source data into correctly formatted, named, and ready-to-submit output files

What this means for your business

Many business processes end with producing a file: an upload template for your ERP, a report for a stakeholder, a formatted extract for a regulator. When that file must be built manually, every column mapping, date format, naming convention, and filter condition is a potential error. A file generation tool reads your data, applies the correct transformations, and produces output files that are named, formatted, and ready to use.

Example: I built a template filler that maps source columns to any upload template using configurable rules (copy, static value, conditional logic, lookups), supports splitting output by group value, and auto-versions files so previous runs are never overwritten.

If you implement this

- A process that involves copying data into 3-4 different templates manually is replaced by a single click that produces all output files at once.
- File naming, date formatting, and column ordering are handled by the tool. The output is consistent every run, regardless of who runs it.
- When your upload template changes, the column mapping is updated in a configuration screen. No code changes, no rebuilding from scratch.

Email Automation & Audit Logging

Send grouped, templated emails from data with a full audit trail

What this means for your business

Follow-up emails, payment notifications, and verification requests often require composing dozens of individual messages with slightly different data each time. An email tool reads your data table, groups related rows by recipient, selects the right template based on conditions, and opens each email in Outlook pre-built and ready for review. Every action is logged.

Example: I built a tool that groups invoice rows by payee, automatically selects between direct, manager escalation, and bank verification templates based on available data, and opens each email as an Outlook draft. A batch of 40 emails that took an hour to compose manually runs in under 2 minutes.

If you implement this

- Batch communications that take an hour of manual composition are reduced to a 2-minute automated run, with each email pre-built, pre-addressed, and ready for your review before sending.
- Template routing means the right message goes to the right person automatically. No more sending the wrong template or forgetting to escalate when an email address is missing.
- A timestamped log records every email sent, skipped, or errored, giving you a clear audit trail for compliance without maintaining a separate tracking spreadsheet.

How I Work

These capabilities are not theoretical. They come from tools I have built and maintained in production for finance and accounting teams. Every engagement starts with understanding your specific workflow and tailoring the solution to it.

1	**Understand the process**	I review the manual workflow step by step to identify where time is lost, where errors enter, and what the output needs to look like.
2	**Build and deliver**	The tool is built in Excel VBA or Python depending on the environment. Tested against real data. Delivered as a self-contained workbook or script.
3	**Document for your team**	Every tool comes with a user manual written for non-technical users. Your team can run it independently from day one without my involvement.
4	**Adapt when things change**	Column mappings, templates, business rules, and file formats are configurable. When requirements change, the tool adapts without a rebuild.

Flat rate pricing. Scope and cost are agreed before work begins. No hourly billing, no surprises.

If any of this sounds like a process your team is currently handling manually, I am happy to discuss what a solution would look like.

No obligation. Send a sample file or describe the process, and I will outline what the tool would do and what it would cost.

Darek Krawczynski

ytcodes.work@gmail.com

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